Diskrete Mathematik

Woche 6 - HG G 26.5

https://github.com/tschai-yim/eth-resources

Feedback (anonym): https://forms.gle/xnkcjsdj86LK6yMh7



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Heutige Agenda

1. Teil (45 min)
- 30 min: Theorie-Übersicht mit Beispielen
- 15 min: Übungen

10 min Pause (hier persönliche Fragen)

2. Teil (45 min): Übungen

Theorie-Übersicht

Kapitel 3.1-3.5

Set Theory (recap)

- **Formalization** like Propositional and Predicate Logic.
- Informality breaks things $\rightarrow$ **Russell's Paradox**.
- Foundation for **everything**.
  - Can build cool stuff from functions to Turing Machines!
- Theoretically starts **with 9 Axioms**.
  - Only applications are important for us.

Set Theory: Basic Concepts (recap)

- **Set and Element ($\in$):**
  - Universe consists only of **sets**; elements are also sets.
  - **Elementhood predicate**: $x \in A$ ("x is an element of A").
- **Equality ($=$):**
  - **Axiom:** sets are equal $\Leftrightarrow$ they have the same elements.
  - Order and duplicates do not matter, e.g., $\{1, 2, 1\} = \{2, 1\}$.
  - **Core Lemma 3.2:** $A = B \Leftrightarrow (A \subseteq B \land B \subseteq A)$.
- **Subsets ($A \subseteq B$):**
  - Every element of A is also an element of B.
  - **Transitivity:** If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$. [Lem 3.3]
- **The Empty Set ($\varnothing$ or $\{\}$):**
  - The **unique** set containing no elements. (Existence from Axiom)
  - It is a subset of every set: $\forall A(\varnothing \subseteq A)$.
- **Example:** $C = \{\varnothing\}$

Constructing and Operating on Sets (recap)

- **Construction:** Theoretically must prove a set exists.
  - Axioms guarantee **singleton sets** $\{a\}$ and finite sets $\{a, b, c\}$.
- **Operations:**
  - **Union ($A \cup B$):** elements in A, or in B, or in both.
  - **Intersection ($A \cap B$):** elements in both A and B.
  - **Difference ($B \setminus A$):** elements in B but not in A.
- **Power Set ($P(A)$):** set of **all subsets** of A.
  - Cardinality: If $|A| = k$, then $|P(A)| = 2^k$.
- **Cartesian Product ($A \times B$):** ordered pairs (a, b) with $a \in A$ and $b \in B$.
  - Cardinality: $|A \times B| = |A| \cdot |B|$.
  - Pair Definition: $(a, b) \stackrel{\text{def}}{=} \{\{a\}, \{a, b\}\}$, ensuring $(a, b) = (c, d) \Leftrightarrow a = c \wedge b = d$.

Relations: Core Concepts

- **Relation ρ from A to B:** Subset of $A \times B$. [Def 3.9]
  - **Notation:** $a \rho b$ instead of $(a, b) \in \rho$.
  - **Example:** $\{(1, c), (2, a), (2, b), (3, b)\}$
  - Like directed graph if relation on A.
- **Identity relation (id_A):** All pairs (a, a) for $a \in A$. [Def 3.10]

Relations: Operations

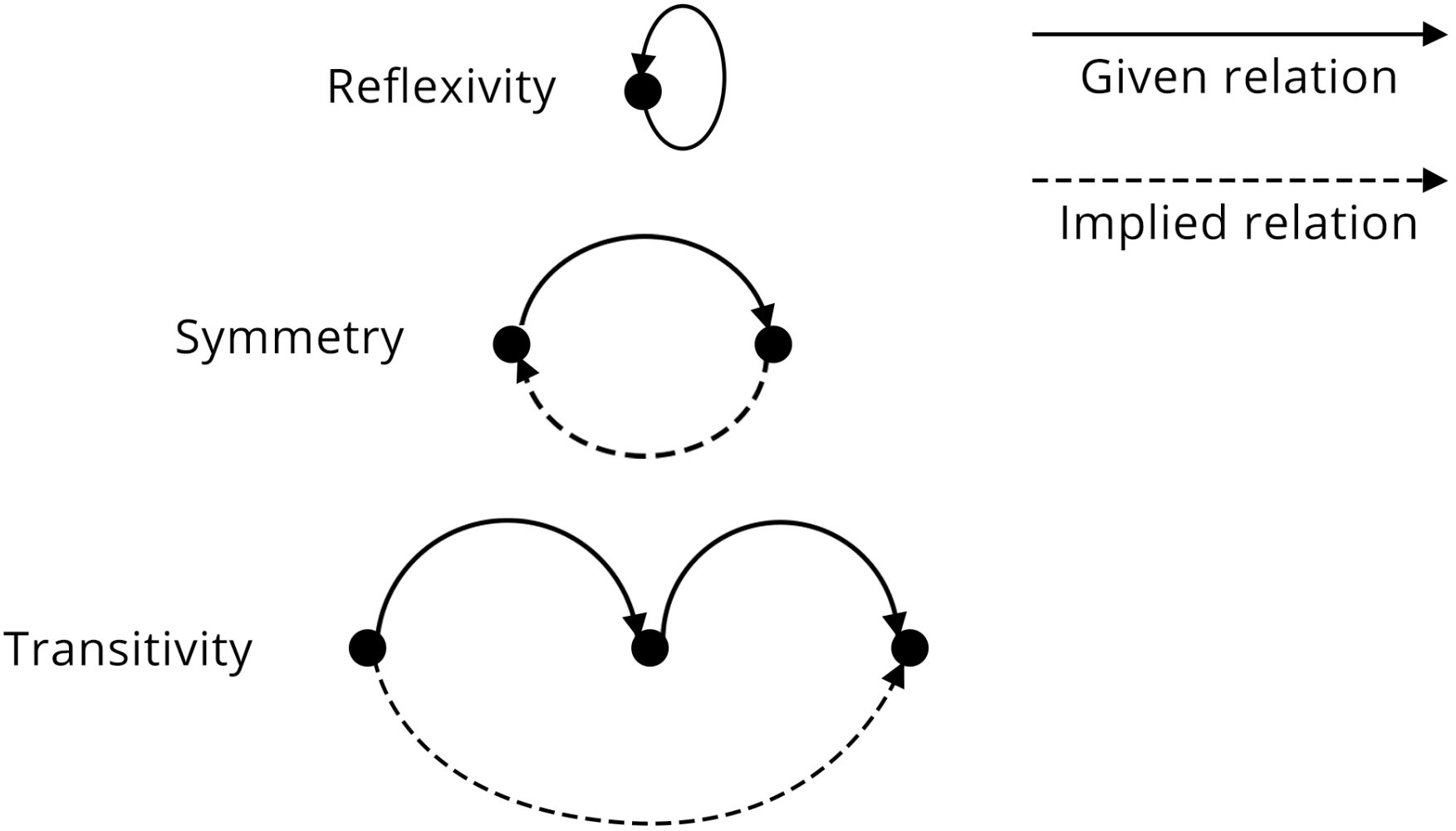
- **Inverse ($\hat{\rho}$ or ρ^{-1}):** $(b, a) \in \hat{\rho}$ for all $(a, b) \in \rho$ [Def 3.11]
  - Intuition: All arrows reversed.
  - Example: Inverse of `child` is `parent`; inverse of $\leq$ is $\geq$.
- **Composition ($\rho \circ \sigma$ or $\rho\sigma$):** (a, c) if $b \in B$ s.t. $a\rho b$ and $b\sigma c$.
  - **Associative**: $(\rho \circ \sigma) \circ \varphi = \rho \circ (\sigma \circ \varphi)$. [Lem 3.7]
  - Inverse **reverses order**: $\widehat{\rho \circ \sigma} = \hat{\sigma} \circ \hat{\rho}$. [Lem 3.8]
  - Example: $\widehat{\text{takes} \circ \text{teaches}}$ connects students to their lecturers.
- **Powers (ρ^n):** n-fold composition of ρ with itself.
  - Example: If parent is a relation, parent^2 is the grandparent relation.
- **Transitive Closure (ρ^*):**
  - Union of all powers: $\rho^* = \bigcup_{n \in \mathbb{N} \setminus \{0\}} \rho^n = \rho \cup \rho^2 \cup \rho^3 \cup \ldots$. [Def 3.18]
  - Represents **reachability**: $a\rho^* b \Leftrightarrow$ path of finite length exists from a to b.
  - Example: The transitive closure of the `parent` relation is the `ancestor` relation.

Relations: Properties

Property	Formal Definition	Set-Theory
Reflexive	$\forall a : a \rho a$ [Def 3.13]	$\mathrm{id} \subseteq \rho$
Irreflexive	$\forall a : \neg(a \rho a)$ [Def 3.14]	$\mathrm{id} \cap \rho = \varnothing$
Symmetric	$\forall a, b : a \rho b \Leftrightarrow b \rho a$ [Def 3.15]	$\rho = \hat{\rho}$
Antisymmetric	$\forall a, b : (a \rho b \wedge b \rho a) \Rightarrow a = b$ [Def 3.16]	$\rho \cap \hat{\rho} \subseteq \mathrm{id}$
Transitive	$\forall a, b, c : (a \rho b \wedge b \rho c) \Rightarrow a \rho c$ [Def 3.17]	$\rho^2 \subseteq \rho$ [Lem 3.9]

- Crucial Distinctions:
  - **Irreflexive $\neq$ Not Reflexive:** no el. self-related vs. one el. not self-related
  - Antisymmetric **not negation** of symmetric.

Relations: Properties



Equivalence Relations

- **Equivalence Relation** is **reflexive**, **symmetric**, and **transitive**. [Def 3.19]
- **Equivalence Class ($[a]_\theta$):**
  - Set of all equivalent to a under θ. [Def 3.20]
  - Notation: $[a]_\theta = \{b \in A \mid b\theta a\}$.
- **Partitioning:**
  - Equivalence rel. **partitions set A** into equivalence classes. [Thm 3.11]
  - Classes **mutually disjoint** and union is A. [Def 3.21]
  - Any two classes are either **identical** ($[a] = [b]$) or **disjoint** ($[a] \cap [b] = \varnothing$).
  - **Quotient set** $\frac{A}{\theta}$ is set of all equivalence classes. [Def 3.22]
- **Intersection ($\cap$):**
  - Of two equivalence rel. is also an equivalence relation. [Lem 3.10]
  - **Example:** The intersection of $\underset{3}{\equiv}$ and $\underset{5}{\equiv}$ on $\mathbb{Z}$ is the relation $\underset{15}{\equiv}$.

Posets: Definition

- **Partial Order** $\preceq$ is **reflexive**, **antisymmetric**, and **transitive**. [Def 3.23]
  - Also called **Order Relation**.
- **Poset (Partially Ordered Set):**
  - Pair $(A; \preceq)$ of set A and a partial order $\preceq$ on it. [Def 3.23]
- Two elements are **comparable** if one is related to the other
  - Otherwise they are **incomparable**. [Def 3.24]
- **Types:**
  - **Partial Order:** *Not all* pairs necessarily comparable.
  - **Total Order:** *All* pairs are comparable (e.g., $(\mathbb{Z}; \leq)$). [Def 3.25]
  - **Well-Ordered:** Total order where every non-empty subset has **least element** (e.g., $(\mathbb{N}; \leq)$). [Def 3.30]
    - Guarantees a "starting point"; crucial for inductive proofs (e.g., Thm 3.17).

Posets: Hasse Diagrams

- Simplified graphical representations for finite posets. [Def 3.27]
- **Covering Relation:** An edge from a up to b exists only if b **covers** a. [Def 3.26]
  - b **covers** a if $a \prec b$ (note $a \neq b$) and no element c exists s.t. $a \prec c \prec b$.

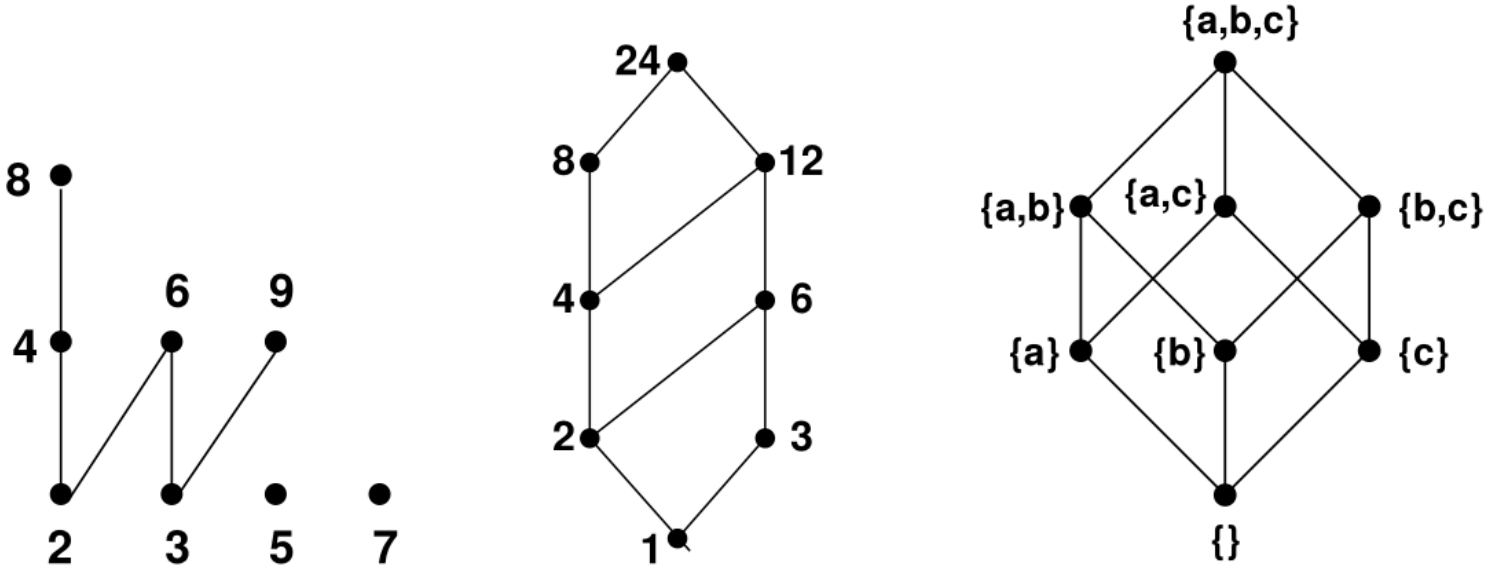


Figure 3.1: The Hasse diagrams of the posets $(\{2,3,4,5,6,7,8,9\};|)$, $(\{1,2,3,4,6,8,12,24\};|)$, and $(\mathcal{P}(\{a,b,c\});\subseteq)$.

Posets: Special Elements

- For the **entire poset A:** [Def 3.29]
  - **Minimal / Maximal:** No element is strictly smaller / larger. (*multiple possible*)
  - **Least / Greatest:** Smaller / larger than or equal to **all** elements. (*unique*)
  - **Difference:** A **least** el. is below *everything*; a **min** el. has *nothing below it.*
- For a **subset $S \subseteq A$:** [Def 3.29]
  - **Lower / Upper Bound:** Element in A (not necessarily in S) that is $\leq$ / $\geq$ all elements in S.
  - **Infimum (glb) / Supremum (lub):** greatest lower bound / least upper bound.
    - Also called **meet** ($\wedge$) / **join** ($\vee$). [Def 3.31]
    - Example $(\mathbb{N}^+; |)$: $12 \wedge 18 = \gcd(12, 18) = 6$.
- **Connection:**
  - Least element of $A = \inf(A)$
  - Greatest element of $A = \sup(A)$.
  - Inf/sup generalize least/greatest to arbitrary subsets.

Posets: Lattices & Combinations

- **Lattices:**
  - A poset where every pair has a **meet** and a **join**. [Def 3.32]
  - Examples:
    - $(\mathcal{P}(A); \subseteq)$, where meet is $\cap$ and join is $\cup$.
    - $(\mathbb{Z} \times \mathbb{Z}, \leq)$, where meet is compo. min and join is comp. max.
- **Combining Posets on $(A \times B)$:**
  - **Direct Product:** [Def 3.28, Thm 3.12]

$$(a_1, b_1) \leq (a_2, b_2) \Leftrightarrow a_1 \preceq a_2 \wedge b_1 \sqsubseteq b_2$$

  - **Lexicographical Order:** dictionary order [Thm 3.13]

$$(a_1, b_1) \underset{\text{lex}}{\leq} (a_2, b_2) \Leftrightarrow (a_1 \prec a_2) \vee (a_1 = a_2 \wedge b_1 \sqsubseteq b_2)$$

Übungen

https://github.com/tschai-yim/eth-resources unter
`03_TA/26HS_252-0025-01L_Diskrete_Mathematik/exercises`

Offene Fragen & Feedback

- Gibt es noch Fragen zu den heutigen Themen oder den Übungen?
- Habt ihr Feedback zur heutigen Übungsstunde?
  - Anonymes Formular: https://forms.gle/xnkcjsdj86LK6yMh7
- Bei Fragen: Schreibt mir eine E-Mail oder auf WhatsApp.

Schöne Woche und bis nächsten Montag!